

# Surface hydroxyl-induced Pt<sup>0</sup> clusters on TiO<sub>2</sub> for synergistic water gas shift catalysis

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The regulation of active site electronic structure is a central strategy for optimizing supported metal catalysts, yet achieving precise and comprehensive control remains challenging due to the complex nature of metal-support interactions. This work presents a practical and reproducible approach to this challenge through the deliberate regulation of surface hydroxyls on TiO<sub>2</sub> via a titanate-derived synthesis and controlled calcination, which correlates with improved water gas shift catalysis activity of Pt/TiO<sub>2</sub> catalyst. The mechanistic interpretation by multiple characterizations suggest that a higher hydroxyl concentration is associated with the formation of more reduced Pt species, which enhanced CO activation. Subsequently, hydroxyl groups can be directly consumed by CO, which shows a correlation with the formation of oxygen vacancies. Catalysts with a higher density of oxygen vacancies exhibit stronger H<sub>2</sub>O dissociation capability. This work develops hydroxyl-content control as a viable strategy for catalyst optimization, supported by a mechanistic framework that rationalizes its impact on both metal electronic structure and oxide surface chemistry.

The water gas shift (WGS) reaction is an important industrial process for hydrogen production and carbon monoxide elimination, where the efficient activation of reactant molecules on the active sites of the catalyst surface is the key determinant of catalytic performance<sup>1</sup>. Supported metal-oxide catalysts have garnered attention due to their high dispersion and stability of metals. On reducible oxide supports, active metal–oxygen vacancies synergy is one possible mechanistic motif for WGS reaction<sup>2,3</sup>. Pt/TiO<sub>2</sub> stands as an exemplary catalytic system that has been extensively investigated in a variety of pivotal reactions, including the WGS reaction, CO oxidation, hydrogenation, and hydrocarbon reforming<sup>4–7</sup>. Currently, due to the electronic metal-support interaction (EMSI) between Pt and TiO<sub>2</sub>, electrons at the interface tend to transfer from Pt species to TiO<sub>2</sub>, resulting in the formation of Pt<sup>δ+</sup>-O<sub>v</sub>-Ti<sup>3+</sup> interfacial sites<sup>8–11</sup>. Although such interfacial sites possess unsaturated Ti<sup>3+</sup> and oxygen vacancies that facilitate the H<sub>2</sub>O dissociation<sup>12–14</sup>, which frequently is the rate-determining step of the WGS reaction<sup>15,16</sup>, the oxidized Pt<sup>δ+</sup> species exhibit limited ability for

CO activation. This limitation hinders effective matching with the enhanced H<sub>2</sub>O dissociation capability, thus restricting the WGS reaction performance. Generally, the enrichment of electron density on Pt sites facilitates the adsorption of CO molecules<sup>2,17</sup>. Therefore, the targeted construction of metallic Pt sites to match the abundant defect structure on the TiO<sub>2</sub> surface represents an effective strategy for further enhancing the WGS reaction, which has been almost overlooked.

Commonly, the electronic structure of metal species is intrinsically linked to their size. The aggregation of metallic Pt particles is often accompanied by the decrease in the dispersion, which, due to geometric effects, results in the loss of active sites<sup>18</sup>. Conversely, ultra-small species (<1 nm) typically exhibit positively charged metal sites, which lead to weakened CO adsorption<sup>19</sup>. Therefore, precise regulation of the electronic structure of Pt sites while controlling the size of metal species is essential yet challenging. It has been reported that the structure of the active metals is closely related to the surface properties of oxide support, which can be modulated by introducing the

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functional groups onto the support<sup>11,20</sup>. Surface hydroxyl groups have been well established for their roles in regulating catalyst structure, influencing the adsorption of reactant molecules, and participating in reactions<sup>21–23</sup>. Recent research has reached a consensus on the use of surface hydroxyl groups constructed on TiO<sub>2</sub> as direct anchoring sites for active metals, which contributes to maintaining the size of the active metals<sup>24–27</sup>. Moreover, a perspective that surface hydroxyl groups on TiO<sub>2</sub> facilitate hydrogen spillover from metal sites to the support surface has been clearly proposed<sup>28,29</sup>. During the H<sup>+</sup> transfer process, electrons are retained at the metal sites<sup>30,31</sup>. Consequently, it can be expected to the correlation between the construction of surface hydroxyl groups on TiO<sub>2</sub> support and the enhancement of electron density at Pt sites. Furthermore, the hydroxyl groups, serving as the crucial intermediates in the WGS reaction<sup>7,32</sup>, are capable of directly reacting with CO adsorbed on Pt<sup>0</sup> sites. This consumption is anticipated to generate the high density of defect sites throughout the catalytic process, thereby facilitating the activation and dissociation of more H<sub>2</sub>O molecules. However, the effectiveness of hydroxyl groups does not always exhibit the positive correlation with their abundance. This is attributed not only to the blocking of metal sites by excess hydroxyl groups, but also to the fact that there exists an optimal amount of hydroxyl groups for modulating the state of the active species<sup>33,34</sup>. Therefore, precise modulation of the hydroxyl content is essential for achieving optimal catalytic performance<sup>35,36</sup>.

In this work, we developed a practical and reproducible hydroxyl-regulation approach via controlled calcination of a titanate precursor to tailor the structure of Pt/TiO<sub>2</sub> catalysts. This method enabled the formation of small Pt<sup>0</sup> clusters (~2 nm) on the hydroxyl-tuned TiO<sub>2</sub> surface, distinct from conventional common Pt<sup>δ+</sup>-O<sub>v</sub>-Ti<sup>3+</sup> interfacial structure. Multiple characterization indicated that the resulting Pt species was consistent with the more reductive electronic state, which enhanced CO adsorption and activation. Furthermore, hydroxyl groups were consumed under CO exposure, a process that correlated with the generation of oxygen vacancies. Catalysts with a higher density of oxygen vacancies exhibited stronger H<sub>2</sub>O dissociation capability. A reasonable mechanistic interpretation was that the synergistic combination of the strong CO adsorption on Pt<sup>0</sup> and efficient H<sub>2</sub>O dissociation enabled the excellent reaction rate of 62.7 mol<sub>CO</sub>·g<sub>Pt</sub><sup>-1</sup>·h<sup>-1</sup> and the turnover frequency (TOF) of 6.15 s<sup>-1</sup> at 250 °C in WGS reaction, which were nearly 2–3 times greater than the Pt/TiO<sub>2</sub> catalysts in the literature. This work thus offered a practical handle for tuning catalytic activity via hydroxyl regulation, alongside insights into how such tuning modulated the metal electronic and oxide surface structure.

## Results

### Regulation of hydroxyl content for the formation of target Pt/TiO<sub>2</sub> catalysts

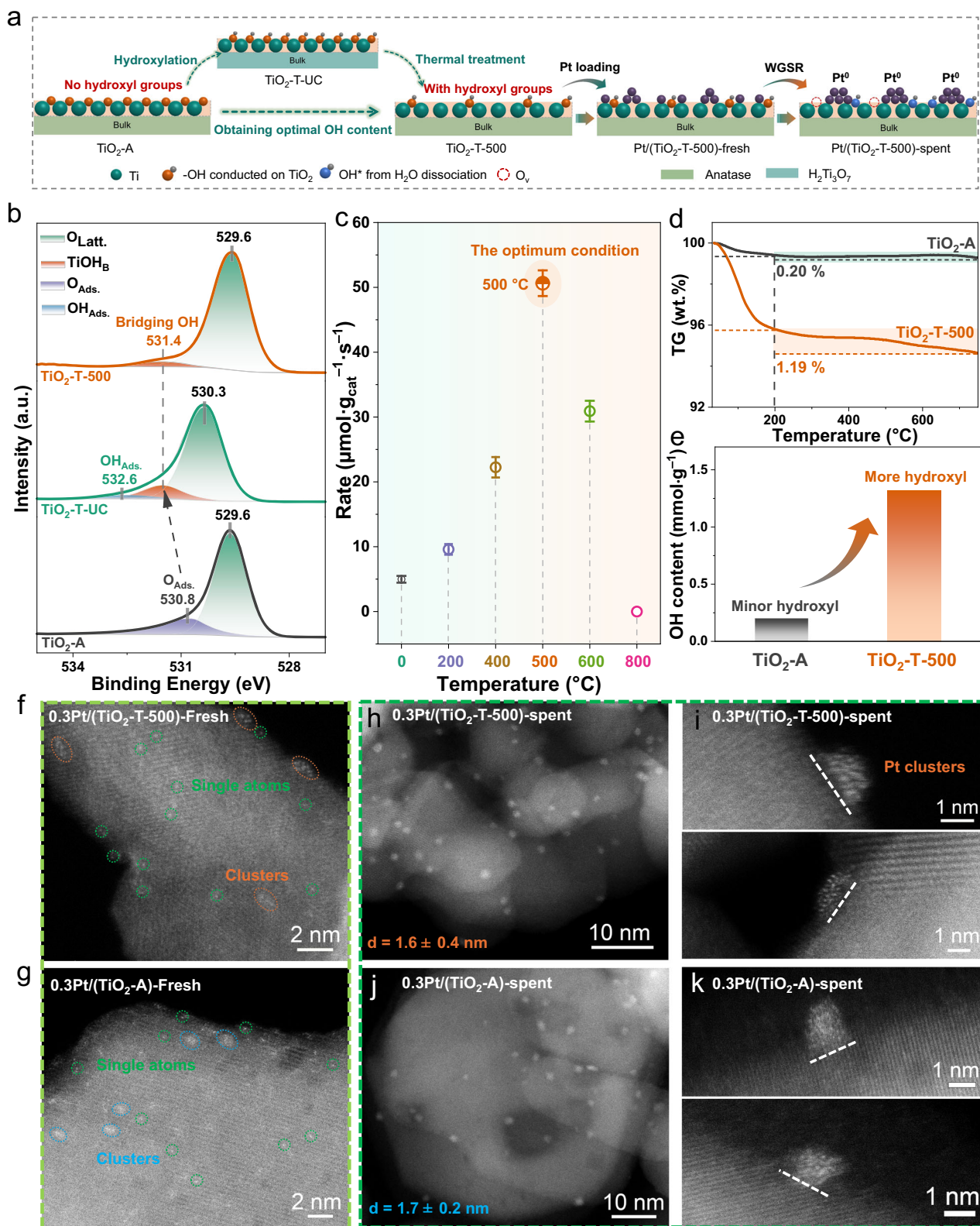
The scheme of regulating the hydroxyl content on the support to obtain the target Pt/TiO<sub>2</sub> catalyst is shown in Fig. 1a. Commercial anatase (denoted as TiO<sub>2</sub>-A) was hydrothermally treated with the highly concentrated NaOH solution (10 mol/L), followed by washing with HCl solution. This process resulted in the phase transformation from anatase to H<sub>2</sub>Ti<sub>3</sub>O<sub>7</sub>, which was denoted as TiO<sub>2</sub>-T-UC (Supplementary Fig. S1). It was imperative to elucidate the surface structure of the modified TiO<sub>2</sub> and the chemical state of the supported Pt species. Firstly, according to the O 1s XPS spectra compared to that of TiO<sub>2</sub>-A, the peak of lattice oxygen in TiO<sub>2</sub>-T-UC shifted toward the higher binding energy, which originated from the distinct layered titanate structure of the H<sub>2</sub>Ti<sub>3</sub>O<sub>7</sub> phase compared to the anatase phase (Fig. 1b)<sup>37</sup>. Moreover, the another peak at 531.4 eV was attributed to bridging hydroxyls and the weak peak appearing at 532.6 eV was considered to be adsorbed hydroxyl groups from the environment, which originated from the absence of any thermal treatment<sup>38</sup>. The results of ATR-IR and TGA experiments also confirmed the formation

of abundant surface hydroxyls (Supplementary Fig. S2). However, the nanotubular morphology of TiO<sub>2</sub>-T-UC caused the supported Pt species to be completely embedded in the TiO<sub>2</sub> support lattice during the WGS reaction (Supplementary Fig. S3). The observed hindrance of CO molecule adsorption and the suppression of activity also confirmed this encapsulation (Supplementary Fig. S4, S5 and Table S1). Therefore, further thermal treatment of TiO<sub>2</sub>-T-UC in the air atmosphere was conducted (Supplementary Fig. S6). As the calcination temperature increased, the nanotubular morphology of TiO<sub>2</sub> progressively collapsed, leading to the exposure of previously encapsulated Pt species (Supplementary Fig. S7 and S8). Concurrently, the hydroxyl concentration decreased with rising calcination temperature, resulting in the increase in the size of the supported Pt species (Supplementary Fig. S9 and S10). By integrating the hydroxyl group content with the exposure and size of Pt species, the Pt-based catalyst supported on TiO<sub>2</sub>-T calcined at 500 °C (denoted as TiO<sub>2</sub>-T-500) exhibited the optimal catalytic activity (Fig. 1c). It was worth noting that O 1s XPS spectra of TiO<sub>2</sub>-T-500 showed that adsorbed hydroxyl groups disappeared due to thermal treatment, but the residual bridging hydroxyls were still present (Fig. 1b). Moreover, the mass loss from hydroxyls desorption of TiO<sub>2</sub>-T-500 in the TGA experiments was 1.19 %, while that of TiO<sub>2</sub>-A was only 0.20 % (Fig. 1d), demonstrating that the TiO<sub>2</sub>-T-500 support possessed the higher abundance of surface hydroxyl groups compared to TiO<sub>2</sub>-A (Fig. 1e)<sup>35</sup>. The H–D exchange and ATR-IR experiment also indicated the hydroxyl groups on TiO<sub>2</sub>-T-500 were more abundant than those on TiO<sub>2</sub>-A (Supplementary Fig. S11). These results revealed that TiO<sub>2</sub>-T-500 support with optimal hydroxyls content was obtained through the two-step hydroxylation-thermal treatment process of TiO<sub>2</sub>-A.

As reported, the surface properties of the support are closely related to the state of the supported active metals. The aberration-corrected HAADF-STEM characterization revealed that in the fresh 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst, Pt species primarily coexisted as single atoms and clusters, which was consistent with the state of Pt species in the fresh 0.3Pt/(TiO<sub>2</sub>-A) catalyst (Fig. 1f, g). After the WGS reaction, the aggregation of Pt species into clusters occurred in both catalysts (Fig. 1h–k and Supplementary Fig. S12 and S13). Statistical analysis of Pt species indicated that the average size in the spent 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst was 1.6 nm, which was quite similar to the 1.7 nm observed in the spent 0.3Pt/(TiO<sub>2</sub>-A) catalyst (Supplementary Fig. S14). Moreover, CO chemisorption measurements revealed that the dispersions of the spent 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) catalysts were 55.2% and 48.1%, respectively (Supplementary Table S1). The non-identical dispersions were derived from CO uptake based on a 1:1 CO:Pt stoichiometry, which might be influenced by various factors, such as support-dependent spillover, dynamic readsorption, or differences in CO binding strength. Nevertheless, they served as a supplementary indication that the Pt clusters in both catalysts remained accessible after reaction. This assessment was consistent with the direct HAADF-STEM observation of similar mean particle sizes after reaction. However, the predominant Pt<sup>0</sup> species were present in the spent 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst, while the Pt<sup>δ+</sup> species were predominant in the spent 0.3Pt/(TiO<sub>2</sub>-A) catalyst (Supplementary Fig. S4). This observation suggested that the hydroxyl groups constructed on the support induced the significant changes in the electronic structure of Pt species in the absence of size effects.

### The formation of Pt<sup>0</sup> species induced by hydroxyls for enhanced CO activation

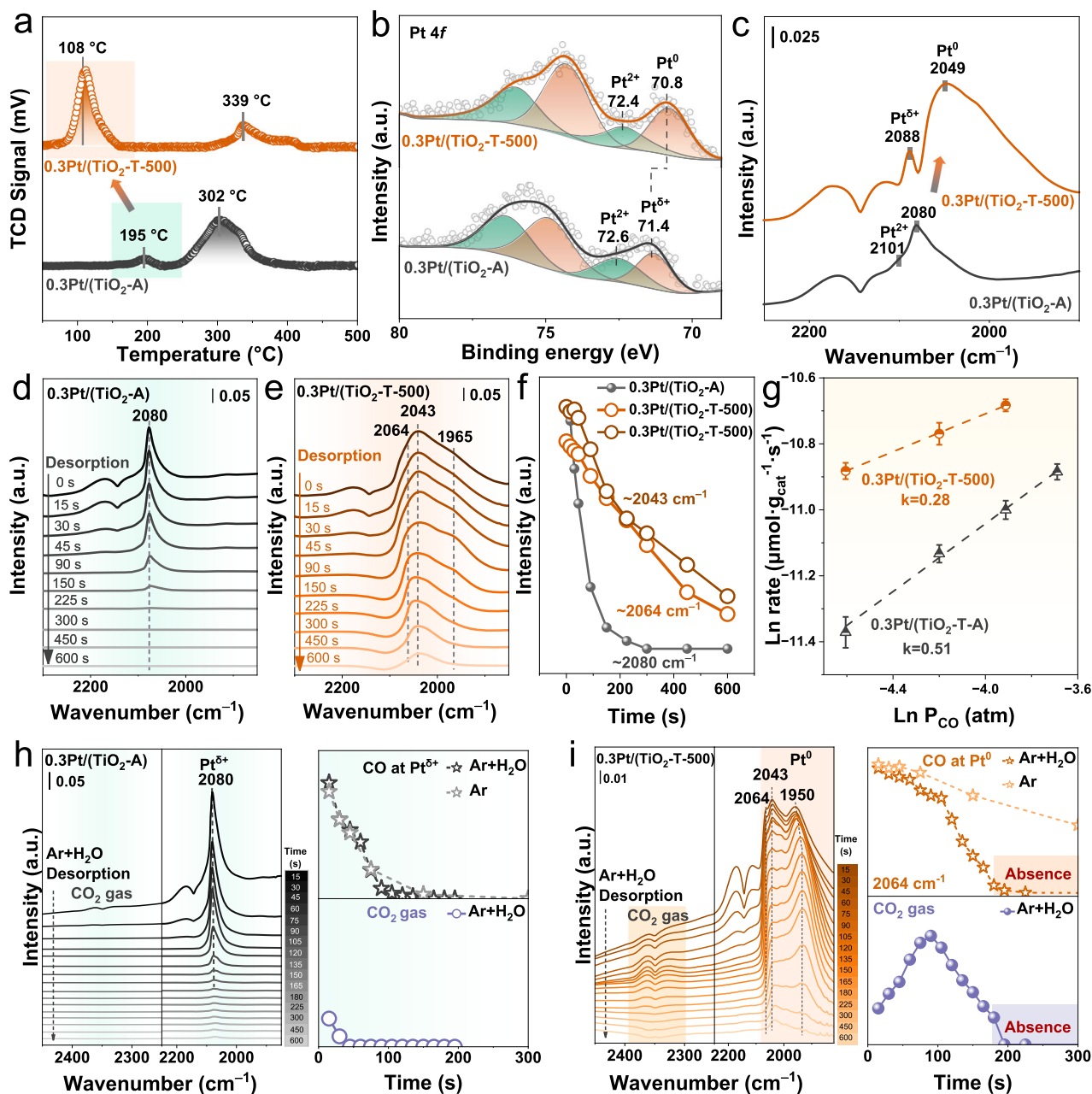
To thoroughly investigate the influence of hydroxyls on the electronic structure of active metals, the H<sub>2</sub> pretreatment process prior to the reaction was taken into consideration, which might induce changes in the catalyst structure. Having ruled out the influence of H<sub>2</sub> reduction temperature on hydroxyl groups and H<sub>2</sub>O desorption (Supplementary Fig. S15a, b), the temperature-programmed reduction by H<sub>2</sub> (H<sub>2</sub>-TPR,



**Fig. 1 | Construction of Pt/TiO<sub>2</sub> catalysts with the optimal hydroxyls content on the support.** **a** Scheme of regulating the hydroxyl content on the support to obtain the target Pt/TiO<sub>2</sub> catalyst. **b** O 1s XPS spectra of TiO<sub>2</sub>-A, TiO<sub>2</sub>-T-UC and TiO<sub>2</sub>-T-500 supports. **c** WGS reaction rate of Pt/TiO<sub>2</sub> catalyst at 250  $^{\circ}\text{C}$  as the function of calcination temperature of TiO<sub>2</sub>-T support, with the optimum at 500  $^{\circ}\text{C}$ . Reaction condition: 2% CO/10% H<sub>2</sub>O/N<sub>2</sub>. The error bars are defined as standard deviation, and

the center of each error bar represents the mean value of three independent experiments. **d** TGA curves of TiO<sub>2</sub>-A and TiO<sub>2</sub>-T-500 supports. **e** The hydroxyl content of TiO<sub>2</sub>-A and TiO<sub>2</sub>-T-500 supports determined from TGA measurements. Aberration-corrected HAADF-STEM images of the fresh (**f**)  $0.3\text{Pt}/(\text{TiO}_2\text{-T-500})$  and (**g**)  $0.3\text{Pt}/(\text{TiO}_2\text{-A})$  catalyst. Aberration-corrected HAADF-STEM images of the spent (**h**, **i**)  $0.3\text{Pt}/(\text{TiO}_2\text{-T-500})$  and (**j**, **k**)  $0.3\text{Pt}/(\text{TiO}_2\text{-A})$  catalyst.





**Fig. 2 | Surface hydroxyl-induced Pt<sup>0</sup> for enhanced CO activation.** **a** H<sub>2</sub>-TPR profiles of 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) samples. **b** Ex situ Pt 4f XPS for 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) samples after H<sub>2</sub> pretreatment. **c** In situ DRIFTS spectra of the CO adsorption for 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) samples after H<sub>2</sub> pretreatment. In situ DRIFTS spectra after WGS reaction of Ar purging over **(d)** 0.3Pt/(TiO<sub>2</sub>-A) and **(e)** 0.3Pt/(TiO<sub>2</sub>-T-500) at 250 °C. **f** The intensity

of CO adsorption bands with the function of time for **(d, e)**. **g** Kinetic orders of CO at 250 °C of 0.3Pt/(TiO<sub>2</sub>-A) and 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst. The error bars are defined as standard deviation, and the center of each error bar represents the mean value of three independent experiments. In situ DRIFTS spectra of Ar+H<sub>2</sub>O purging over **(h)** 0.3Pt/(TiO<sub>2</sub>-A) and **(i)** 0.3Pt/(TiO<sub>2</sub>-T-500) at 250 °C. Right: the intensity of CO adsorption on Pt sites upon Ar+H<sub>2</sub>O and Ar purging as function of time.

Fig. 2a) showed that the significant decrease in the reduction temperature of the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst compared to 0.3Pt/(TiO<sub>2</sub>-A). This elucidated that the construction of surface hydroxyls on the TiO<sub>2</sub>-T-500 support was beneficial to the reduction of catalyst, which was further evidenced by the higher hydrogen consumption observed in 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst (Supplementary Fig. S15c). According to the Pt 4f XPS spectra, the Pt species in the fresh 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) catalysts exhibited Pt<sup>2+</sup> species at 72.5 eV and 72.3 eV, respectively (Supplementary Fig. S16)<sup>39,40</sup>. After H<sub>2</sub> pretreatment, the Pt in the 0.3Pt/(TiO<sub>2</sub>-A) catalyst showed the binding energy of 71.4 eV, corresponding to Pt<sup>6+</sup> (Fig. 2b)<sup>41,42</sup>. In contrast, the binding energy for

the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst shifted down by 0.7 eV to 70.8 eV, which likely correlated with the presence of Pt<sup>0</sup><sup>43</sup>. This result was consistent with the superior reducibility of the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst compared to 0.3Pt/(TiO<sub>2</sub>-A), facilitating the reduction of Pt<sup>2+</sup> to Pt<sup>0</sup> during H<sub>2</sub> treatment. Furthermore, the XANES profile at Pt L<sub>3</sub> edge spectra revealed the lower white-line intensity for 0.3Pt/(TiO<sub>2</sub>-T-500) compared to 0.3Pt/(TiO<sub>2</sub>-A) (Supplementary Fig. S17), which indicated that the Pt species in 0.3Pt/(TiO<sub>2</sub>-T-500) possessed the higher 5d electron density, corresponding to the more Pt<sup>0</sup> sites<sup>44</sup>. However, considering the poor signal-to-noise ratio and interference from H<sup>+</sup> adsorption under H<sub>2</sub> atmosphere, this result could not fully

represent the exact state of Pt species under the WGS reaction, which could only imply a stronger tendency for Pt species in 0.3Pt/(TiO<sub>2</sub>-T-500) to be in the metallic state compared to that in 0.3Pt/(TiO<sub>2</sub>-A). Moreover, in situ DRIFTS experiments with CO adsorption showed that the weak reducibility of 0.3Pt/(TiO<sub>2</sub>-A) catalyst resulted in the presence of Pt<sup>2+</sup> and Pt<sup>δ+</sup> (0 < δ < 2) species, observed at 2101 cm<sup>-1</sup> and 2080 cm<sup>-1</sup>, respectively<sup>45,46</sup>. In contrast, while the Pt<sup>2+</sup> species disappeared in the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst, a new band appeared at 2049 cm<sup>-1</sup>. Its red shift relative to the Pt<sup>δ+</sup> band in 0.3Pt/(TiO<sub>2</sub>-A) was associated with its assignment to Pt<sup>0</sup> species, consistent with literature and electronic effects<sup>46–50</sup>. While we acknowledge that band positions can also be influenced by factors such as particle size and CO coverage, the observed trend across the sample series, combined with the supporting HAADF-STEM images, XPS and XANES data, leads us to attribute the dominant effect to the modification of the Pt electronic state. Overall, the observed trends were consistent with a mechanistic interpretation wherein support hydroxylation promoted the formation of Pt species with the more metallic electronic character.

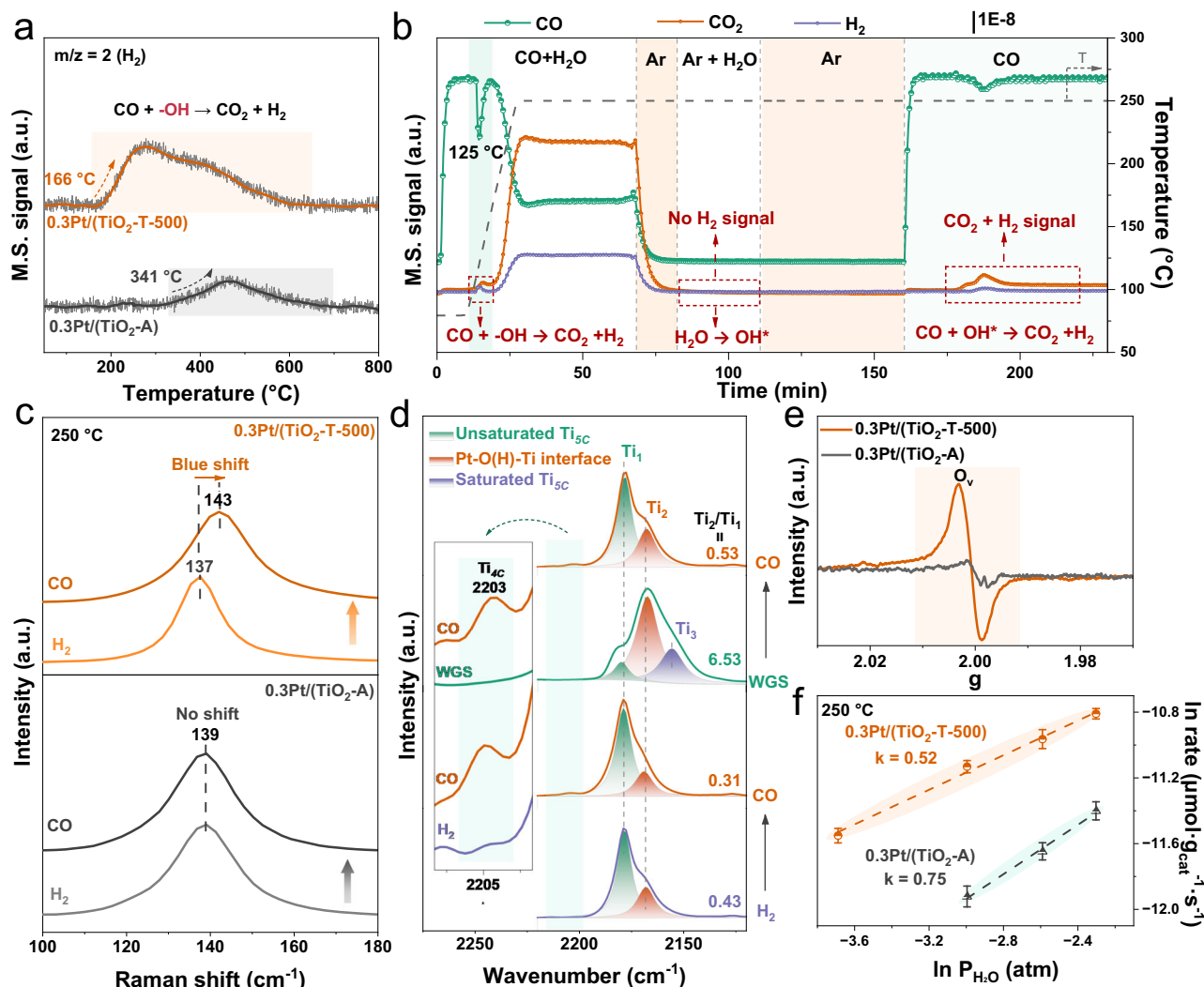
During the subsequent WGS reaction, the active metal typically served as the adsorption and activation site for the reactant molecule CO. In situ DRIFTS experiments revealed the Pt<sup>0</sup> species in 0.3Pt/(TiO<sub>2</sub>-T-500) and the Pt<sup>δ+</sup> species in 0.3Pt/(TiO<sub>2</sub>-A) formed during the H<sub>2</sub> pretreatment, which remained under WGS reaction condition (Supplementary Fig. S18). To explore the adsorption capability of CO molecules on Pt species with different electronic states in the two catalysts, Ar atmosphere was switched from WGS reaction atmosphere in the in situ DRIFTS experiment (Fig. 2d, e). The Pt<sup>δ+</sup> species in 0.3Pt/(TiO<sub>2</sub>-A) catalyst showed limited affinity for CO adsorption, with nearly complete desorption occurring within 3 minutes under Ar gas (Fig. 2d). For 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst, in addition to the bridged CO signal at 1950 cm<sup>-1</sup> originating from the metal-support interface, CO linear adsorption on the masked Pt<sup>0</sup> sites at 2064 cm<sup>-1</sup> emerged during Ar purging, which were closer to the interfacial sites compared to the Pt<sup>0</sup> at 2043 cm<sup>-1</sup> (Fig. 2e)<sup>47,49</sup>. Notably, the interfacial signal at 2064 cm<sup>-1</sup> remained red-shifted compared to the Pt<sup>δ+</sup> signal at 2080 cm<sup>-1</sup> in 0.3Pt/(TiO<sub>2</sub>-A), indicating that the interfacial Pt sites in the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst possessed higher electron density, consistent with the state of Pt<sup>0</sup> clusters. The persistence of CO adsorption strength for the two kinds of Pt<sup>0</sup> species in the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst after 10 min of Ar gas purging indicated that the Pt<sup>0</sup> induced by surface hydroxyls exhibited the stronger CO adsorption capability (Fig. 2f). Moreover, the enhanced binding of CO on the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst was further evidenced by the CO desorption temperature in the CO-TPD experiment (Supplementary Fig. S19), which was 30 °C for 0.3Pt/(TiO<sub>2</sub>-A) and significantly lower than the 65 °C observed for 0.3Pt/(TiO<sub>2</sub>-T-500). Furthermore, the kinetic orders of CO for the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst was 0.07 at 220 °C and 0.27 at 250 °C, which was one-third of the value for the 0.3Pt/(TiO<sub>2</sub>-A) catalyst at 220 °C and half of their value at 250 °C (Fig. 2g and Supplementary Fig. S20). The decrease in the CO reaction order further provided evidence for the significantly enhanced adsorption capability of CO molecules by Pt<sup>0</sup> species in the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst.

It has been reported that the electronic structure of Pt species in the WGS reaction has been a subject of considerable controversy<sup>45,51</sup>. To accurately elucidate the activation capability of CO by Pt species with different electronic structures, in situ DRIFTS experiments were employed at 250 °C, involving switching between CO and Ar or Ar+H<sub>2</sub>O for adsorption and desorption steps. For the 0.3Pt/(TiO<sub>2</sub>-A) catalyst, after eliminating interference from surface oxygen on the gaseous CO<sub>2</sub> signal (Supplementary Fig. S21), Ar and Ar+H<sub>2</sub>O atmosphere was switched from CO adsorption (Fig. 2h and Supplementary Fig. S22). The CO adsorbed on the oxidized Pt<sup>δ+</sup> species of 0.3Pt/(TiO<sub>2</sub>-A) catalyst showed similar desorption rates under both Ar+H<sub>2</sub>O and Ar atmospheres and the concurrent decrease in the gaseous CO<sub>2</sub> signal, which indicated that Pt<sup>δ+</sup> species was not the efficient catalytic site. In

contrast, the CO adsorbed on both the Pt<sup>0</sup> species in the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst at 2064 cm<sup>-1</sup> and at 2043 cm<sup>-1</sup> exhibited the significantly accelerated consumption rate under Ar+H<sub>2</sub>O purging compared to Ar purging, accompanied by the gaseous CO<sub>2</sub> signal that initially increased and subsequently decreased (Fig. 2i and Supplementary Fig. S23). This indicated that CO adsorbed on Pt<sup>0</sup> sites reacted with H<sub>2</sub>O and generated additional CO<sub>2</sub>. As the adsorbed CO on Pt<sup>0</sup> was progressively consumed, the production of CO<sub>2</sub> gradually diminished and eventually vanished upon complete purging. These results collectively pointed to the superior ability of Pt<sup>0</sup> in activating CO molecules. In addition, the CO on the Pt<sup>0</sup> species near the interface at 2064 cm<sup>-1</sup> was completely consumed after 200 s, and correspondingly, CO<sub>2</sub> was no longer generated. Meanwhile, the CO signal on the Pt<sup>0</sup> away from the interface at 2043 cm<sup>-1</sup> remained present, but continued injection of H<sub>2</sub>O could not generate additional CO<sub>2</sub> (Supplementary Fig. S24). These results revealed that the Pt<sup>0</sup> closer to the interface was more active in activating CO molecules.

### The participation of hydroxyls in the reaction to generate oxygen vacancies

Generally, the activation and dissociation of H<sub>2</sub>O molecule was commonly regarded as the rate-determining step in the WGS reaction, which could be substantiated by H/D kinetic isotope effect (KIE) experiments (Supplementary Fig. S26)<sup>7</sup>. In the association mechanisms of the WGS reaction, surface hydroxyls, formed through H<sub>2</sub>O dissociation, served as active intermediates that react with CO to generate products<sup>52</sup>. Therefore, it was crucial to confirm the reactivity of surface hydroxyls constructed on the support of 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst during the reaction. As shown in Fig. 3a, CO-TPR measurement after H<sub>2</sub> treatment was conducted to monitor the H<sub>2</sub> generation. Upon CO injection, the H<sub>2</sub> signal for 0.3Pt/(TiO<sub>2</sub>-T-500) appeared at 166 °C, which was not only lower than the H<sub>2</sub> formation temperature in the 0.3Pt/(TiO<sub>2</sub>-A) catalyst, but also showed the significant increase in the amount of H<sub>2</sub> generated. This results indicated that the surface hydroxyls constructed on the catalyst could be directly consumed by CO to form products. To further elucidate the participation behavior of surface hydroxyls of 0.3Pt/(TiO<sub>2</sub>-T-500) in the reaction, mass spectrometry was performed under different conditions with continuous operation (Fig. 3b). Firstly, the WGS reaction condition was implemented to track the consumption of reactants and the formation of products. At 125 °C, the rapid CO consumption occurred before the reaction initiation, accompanied by the simultaneous formation of CO<sub>2</sub> and H<sub>2</sub>. This phenomenon might originate from the participation of surface hydroxyls on the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst in the reaction, which was further evidenced by the fact that the 0.3Pt/(TiO<sub>2</sub>-A) catalyst exhibited no H<sub>2</sub> production before the reaction initiation (Supplementary Fig. S27). Following the WGS reaction, it was observed that there was an absence of H<sub>2</sub> generation after the injection of H<sub>2</sub>O, which excluded the presence of the redox pathway<sup>52,53</sup>. After a prolonged high-temperature drying process under Ar atmosphere to remove surface moisture, the temperature programmed surface reaction (TPSR) experiment was conducted. As injecting CO, the distinct signals of CO<sub>2</sub> and H<sub>2</sub> with the ratio of 2:1 were exhibited, which indicated the association mechanism of the catalyst following H<sub>2</sub>O dissociation forming hydroxyls as active intermediates (Supplementary Fig. S28a). The similar reaction mechanism was also followed by 0.3Pt/(TiO<sub>2</sub>-A) catalyst (Supplementary Fig. S28b). This further demonstrated the



**Fig. 3 | Surface hydroxyls consumed by CO to facilitate oxygen vacancy formation for enhanced H<sub>2</sub>O dissociation.** **a** CO-TPR profiles of 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) samples after H<sub>2</sub> treatment. **b** In situ WGS reaction, H<sub>2</sub>O dissociation and CO-TPSR of the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst. **c** In situ Raman of 0.3Pt/(TiO<sub>2</sub>-A) and 0.3Pt/(TiO<sub>2</sub>-T-500) during the switch from H<sub>2</sub> to CO at 250 °C. **d** In situ

infrared spectra of 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst recorded after various pretreatment conditions at -130 °C. **e** EPR measurement of 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) after WGS reaction. **f** H<sub>2</sub>O kinetic orders of 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) at 250 °C. The error bars are defined as standard deviation, and the center of each error bar represents the mean value of three independent experiments.

crucial role of surface hydroxyls in the efficient progress of WGS reactions.

Given that the consumption of surface hydroxyls constructed on the catalyst by CO occurred prior to the onset of the reaction, it is imperative to reveal the role of the consumption in surface structure of the catalysts. Firstly, in situ Raman spectroscopy was conducted (Fig. 3c). Upon switching from H<sub>2</sub> to CO, the  $F_{2g}$  peak of the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst exhibited the blue shift of 7 cm<sup>-1</sup>, while that in the 0.3Pt/(TiO<sub>2</sub>-A) catalyst remained unchanged<sup>54,55</sup>. A plausible explanation for this difference was that the reaction between CO and the surface hydroxyls of the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst not only generated the products but also facilitated the creation of oxygen vacancies. In addition, in situ infrared spectroscopy at low temperature (-130 °C) was employed by detecting Ti sites after treatment under various conditions at 300 °C (Fig. 3d). The two peaks located at 2179 cm<sup>-1</sup> and 2164 cm<sup>-1</sup> appeared after H<sub>2</sub> pretreatment, which were attributed to unsaturated Ti<sub>5c</sub> sites (denoted as Ti<sub>1</sub>) and Pt-O(H)-Ti interfacial sites (denoted as Ti<sub>2</sub>), respectively<sup>36,57</sup>. The ratio of the peak areas of Ti<sub>2</sub> to Ti<sub>1</sub> sites (Ti<sub>2</sub>/Ti<sub>1</sub>) was used to qualitatively represent the content of hydroxyls. Subsequently, the catalyst was further treated with CO at

300 °C and then cooled down to -130 °C for testing. It was found that the Ti<sub>2</sub>/Ti<sub>1</sub> ratio decreased from 0.43 to 0.31, indicating that CO consumed the surface hydroxyls. Simultaneously, the peak intensity at 2203 cm<sup>-1</sup>, attributed to Ti<sub>4c</sub> sites, increased significantly, which indicated the increase in oxygen vacancies on the TiO<sub>2</sub> surface<sup>56,57</sup>. The above results provided insight into the correlation between the consumption of surface hydroxyl groups by CO and the facilitated formation of oxygen vacancies. After the WGS reaction, the signal of Pt-O(H)-Ti interfacial sites re-emerged abundantly, which was likely attributable to the surface hydroxyls formed via H<sub>2</sub>O dissociation during the reaction. Consistent with this, the post-reaction O 1s XPS spectra and ATR-IR measurements likewise showed enhanced signals associated with hydroxyl species (Supplementary Fig. S29). Moreover, the weakening signal of the Ti<sub>4c</sub> and unsaturated Ti<sub>5c</sub> sites, along with the formation of new saturated Ti<sub>5c</sub> sites (denoted as Ti<sub>3</sub>)<sup>56,57</sup>, indicated that oxygen vacancies might be replenished by hydroxyls generated from water dissociation. Afterward, the post-reaction catalyst was also treated with CO. The Pt-O(H)-Ti interfacial sites peak weakened and the Ti<sub>4c</sub> and unsaturated Ti<sub>5c</sub> sites peaks increased, which was attributed to the consumption of hydroxyl groups by CO and the concomitant



formation of oxygen vacancies. A reasonable model to explain these observations was that the consumption of hydroxyl groups by CO, whether initially present or formed during the WGS reaction, suggested a link to the formation of oxygen vacancies. Furthermore, the EPR results manifested that the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst did not exhibit significant defects after H<sub>2</sub> pretreatment (Supplementary Fig. S30), while the substantial number of defects emerged following the WGS reaction (Fig. 3e). And the quantity of these defects was significantly higher compared to the 0.3Pt/(TiO<sub>2</sub>-A) catalyst. Given the similar size of the Pt species in both catalysts, the variation in oxygen vacancy concentration could be rationalized by the differing levels of surface hydroxylation, which was consistent with the observed correlation between hydroxyl consumption and defect formation.

Furthermore, insights were gained into the behavior of water dissociation on the surface of the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst using isotopic experiments. The catalyst was first subjected to CO-TPR aimed at removing hydroxyl groups. Subsequently, H<sub>2</sub><sup>18</sup>O was introduced for 20 min. Upon switching back to the CO flow, signals for C<sup>16</sup>O<sup>18</sup>O were detected by both in situ DRIFTS and mass spectrometry (Supplementary Fig. S31 and S32). This observation suggested the possibility that H<sub>2</sub>O dissociated on the catalyst surface. Moreover, 0.3Pt/(TiO<sub>2</sub>-T-500) with more defects exhibited the greater capacity for H<sub>2</sub>O dissociation, as evidenced by the fact that the reaction order of H<sub>2</sub>O at both 250 °C and 290 °C was lower than that of 0.3Pt/(TiO<sub>2</sub>-A) catalyst (Fig. 3f and Supplementary Fig. S33). However, the present experiments unfortunately could not confirm the exact active site for H<sub>2</sub>O dissociation. Moreover, given the dynamic nature of hydroxyl groups under reaction conditions, their precise role could not be unambiguously assigned to individual steps. Taken together, a plausible framework could be inferred by integrating all available data, namely, that the hydroxyl-tuning strategy resulted in an optimal catalyst concurrently exhibiting higher oxygen vacancy signatures and the greater capacity for water activation.

### The enhanced WGS reaction performance by the construction of hydroxyls

The comprehensive optimization of both the electronic structure of the active metal and the surface structure of the support oxide by surface hydroxyl groups has been clearly revealed through the above results, which was summarized in the schematic diagram in Fig. 4a. On one hand, surface hydroxyls facilitated the reduction of the catalyst, thereby forming Pt<sup>0</sup> species that enhanced the adsorption and activation of CO molecules. On the other hand, surface hydroxyls were directly consumed by CO during the reaction, which promoted the formation of oxygen vacancies that enhanced the dissociation of H<sub>2</sub>O molecules. Therefore, in contrast to the singular focus on optimizing the state of Pt species or the TiO<sub>2</sub> surface structure reported in the literature, surface hydroxyls specifically achieved the simultaneous enhancement in the activation capability for both reactant molecules in the WGS reaction, thereby working in combination to maximize the catalytic performance.

The WGS reaction was evaluated under the product free gas (2% CO/10% H<sub>2</sub>O/N<sub>2</sub>). The catalytic activity of 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst was significantly higher than that of 0.3Pt/(TiO<sub>2</sub>-A) catalyst, whose CO<sub>2</sub> conversion approached nearly 100% at 250 °C (Fig. 4b). The combined evidence from ICP-MS quantification, XPS analysis, and thorough support washing ruled out the contribution of residual Na and Cl from synthesis to the observed activity enhancement (Supplementary Fig. S34 and Table S3). The similar enhanced performance was also exhibited by catalysts with Pt loadings of 0.1 wt.% and 0.5 wt.% (Supplementary Fig. S35). Moreover, the WGS reaction activity was improved with increasing Pt loading from 0.1 wt.% to 0.3 wt.% on TiO<sub>2</sub>-T-500, while further increase of Pt loading to 0.5 wt.% led to inferior activity, which might be caused by saturation of the Pt active site (Supplementary Fig. S36). Thus, 0.3Pt/(TiO<sub>2</sub>-T-500) sample turned out

as the ideal catalyst, which established the apparent advantage of reaction rate at 225 °C and 250 °C compared to other catalyst systems in the literature (Fig. 4c and Supplementary Table 4). In particular, the catalytic activity of 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst reached  $1.7 \times 10^4 \mu\text{mol}_{\text{CO}} \cdot \text{g}_{\text{Pt}}^{-1} \cdot \text{s}^{-1}$  at 250 °C, which was more than twice that of the previously reported Pt/TiO<sub>2</sub> catalysts. Moreover, the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst exhibited the TOF nearly twofold higher than that of the reference 0.3Pt/(TiO<sub>2</sub>-A) catalyst and advantages over other Pt-based catalysts reported in the literature (Supplementary Table 5). It should be noted that these value were derived via normalization using the uncertain dispersion estimate, which was subject to the inherent assumptions (1:1 CO:Pt stoichiometry) and potential interference. Assuming that the CO chemisorption measurements were comparable for the both catalysts, the observation of a markedly higher TOF indicated a potentially greater intrinsic activity of the active sites on the TiO<sub>2</sub>-T-500 support. The apparent activation energies (*E<sub>a</sub>*) obtained from the Arrhenius plots showed the similar *E<sub>a</sub>* values of 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) catalysts (Supplementary Fig. S37), which suggested the two samples followed the identical reaction mechanism, consistent with the results of TPSR experiment. Furthermore, the long-term stability test of 0.3Pt/(TiO<sub>2</sub>-T-500) catalysts was examined that there was no deactivation with 70 h, which also possessed remarkable cyclic stability during the rising and cooling between 250 °C and 300 °C (Fig. 4d). Correspondingly, Pt 4f XPS spectra indicated the slight further reduction of the Pt species during the long-time reaction (Supplementary Fig. S38). In contrast, the 0.3Pt/(TiO<sub>2</sub>-A) catalyst exhibited the gradual increase in activity as the reaction progressed, eventually stabilizing after 20 h, although its final activity remained lower than that of the 0.3Pt/(TiO<sub>2</sub>-T-500) catalyst. This phenomenon originated from the enhanced electron density of Pt species, as evidenced by the peak shifting to lower binding energy in the post-stability Pt 4f XPS characterization (Supplementary Fig. S39). This further confirmed the crucial role of inducing Pt<sup>0</sup> formation for the enhancement of catalytic performance.

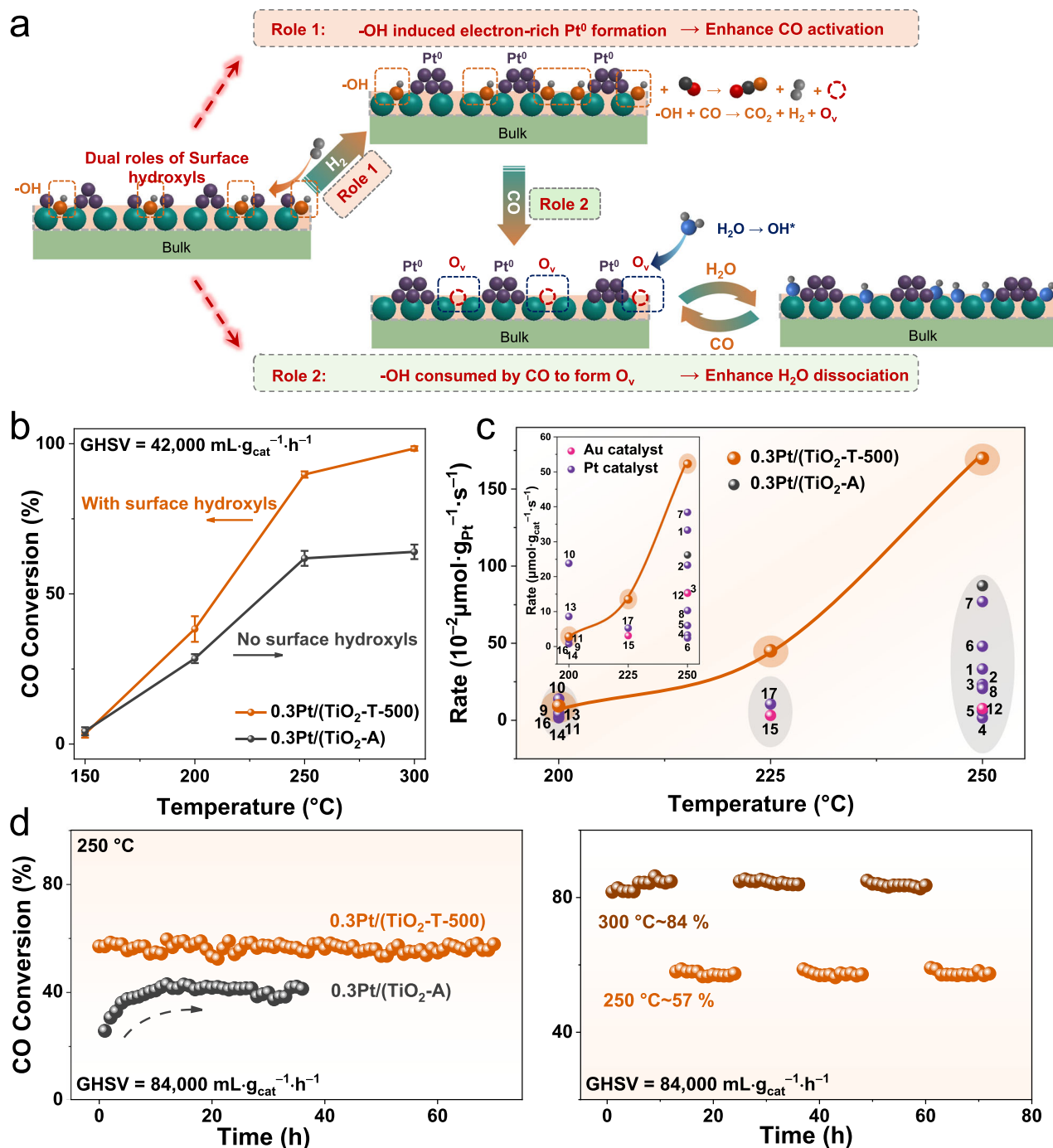
## Discussion

The comprehensive and targeted modulation of catalytic sites is closely related to the adsorption and activation capability of reactant molecules. In this work, a practical and reproducible hydroxyl-regulation approach via controlled calcination of a titanate precursor was developed to form ~2 nm Pt<sup>0</sup> clusters on the hydroxyl-tuned TiO<sub>2</sub> support. The Pt<sup>0</sup> acted as the active sites enhancing the adsorption and activation for CO molecules. Subsequently, surface hydroxyls also directly participated in the reaction to generate products. Through comprehensive in situ characterizations, it was observed that the consumption of hydroxyls by CO during the reaction was consistent with oxygen vacancy formation. The obtained catalysts with higher oxygen vacancies content exhibited stronger H<sub>2</sub>O dissociation capability. The collectively enhanced activation capabilities of both CO and H<sub>2</sub>O molecules might be associated with the improved WGS reaction performance. This work has developed a method for controlling the hydroxyl content to optimize catalytic activity, and has provided a reasonable explanation related to multiple characterizations regarding the regulation of hydroxyl on the metal electronic structure and oxide surface structure.

## Methods

### Preparation of TiO<sub>2</sub> nanotube and subsequent thermal treatment process

The preparation of the TiO<sub>2</sub> tube involved using a hydrothermal technique within Teflon-lined stainless-steel autoclaves<sup>37,38</sup>. In a standard synthesis procedure for the support, 28 g of NaOH (sourced from Sinopharm) was dissolved in 70 mL of deionized water, resulting in a 10 mol/L solution. Subsequently, 2 g of commercial anatase TiO<sub>2</sub> (obtained from Macklin, with the particle size of 25 nm, referred to as



**Fig. 4 | Leveraging the roles of surface hydroxyls to enhance reaction performance.** **a** Scheme of dual roles of surface hydroxyls in optimizing the catalyst structure and participating in the reaction. **b** Temperature-dependent CO conversion of 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) catalysts with 0.3% loading for the WGS reaction. Reaction condition: 2% CO/10% H<sub>2</sub>O/N<sub>2</sub>, GHSV = 42,

$000 \text{ mL} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$ . **c** Comparison of reaction rate with other Pt/TiO<sub>2</sub> catalysts. **d** The stability 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) catalysts. Left: long-term stability at 250 °C of 0.3Pt/(TiO<sub>2</sub>-T-500) and 0.3Pt/(TiO<sub>2</sub>-A) catalysts. Right: the cyclic stability at 300 °C and 250 °C for 0.3Pt/(TiO<sub>2</sub>-T-500) at  $84,000 \text{ mL} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$ .

TiO<sub>2</sub>-A) was introduced into the solution and stirred for 1 h at room temperature. Following this, the mixture was subjected to heating at 130 °C for a duration of 24 h. Post-hydrothermal synthesis, the freshly formed precipitates were isolated by filtration and then washed using deionized water and 2 L of a 0.2 M HCl (Sinopharm) aqueous solution, followed by another round of washing with deionized water. The resultant material was then dried in air at 120 °C for 18 h and subsequently ground using a mortar. From this point forward, the TiO<sub>2</sub> tube

was designated as TiO<sub>2</sub>-T-UC. Subsequently, the TiO<sub>2</sub>-T-UC powder was heated in a tube furnace at a rate of 2 °C/min to 200 °C, 400 °C, 500 °C, 600 °C and 800 °C, respectively, and maintained at each temperature for 8 h. The resulting samples were denoted as TiO<sub>2</sub>-T-t (t = 200, 400, 500, 600, 800), respectively. The TiO<sub>2</sub>-T-wash-500 catalyst was prepared following an identical procedure to that of TiO<sub>2</sub>-T-500, with the modification that after washing with 2 L of HCl, the solid was further washed with 5 L of high-purity water. The reference



commercial anatase was also heated in a tube furnace at a rate of 2 °C/min to 500 °C for 8 hours before use.

### Preparation of various Pt/TiO<sub>2</sub> catalysts

The platinum–titanium catalysts were prepared according to the wet impregnation (IM) method. First, 0.2 g of TiO<sub>2</sub> powders was added to 20 mL high purity water and dispersed uniformly under vigorous stirring. After stirring the mixture for 30 min, a designed amount of chloroplatinic acid (H<sub>2</sub>PtCl<sub>6</sub>) solution (0.19 mol/L) was added into the above slurry. The mixture was heated to 90 °C in an oil bath with continuous stirring to facilitate drying. The obtained catalysts were labeled as xPt/(TiO<sub>2</sub>-A), xPt/(TiO<sub>2</sub>-T-UC) and xPt/(TiO<sub>2</sub>-T-t). The symbol x denotes the weight percentage of metallic Pt on the support, which are 0.1, 0.3, and 0.5 wt%.

### Inductively coupled plasma mass spectrometry (ICP-MS)

ICP-MS analysis was performed using the NexION 1000 G instrument. The solid catalysts were completely digested using the high-concentration hydrofluoric acid (HF). Approximately 4 mg of the solid catalyst powder was completely digested in 2 mL of HF and subsequently diluted to a final volume of 100 mL. A 10 mL aliquot of this solution was then transferred to a new volumetric flask and diluted again to 100 mL. The resulting solution was used for ICP-MS analysis. The instrument detection limit for the ICP-MS was 50 ppt. By converting this value relative to the initial mass of the catalyst, the corresponding detection limit for elemental content in the solid catalyst is 0.00125 wt.%.

### Temperature-programmed reduction with H<sub>2</sub> (H<sub>2</sub>-TPR)

H<sub>2</sub>-TPR experiments were performed using a Builder PCSA-1000 instrument. Prior to the test, the catalysts (30 mg, 20–40 mesh) were pretreated under Ar at 120 °C for 30 min, followed by switching to 5% H<sub>2</sub>/Ar. After the baseline was stabilized in hydrogen, the temperature was increased from room temperature to 500 °C at a rate of 10 °C·min<sup>-1</sup>.

### Temperature programmed reduction with CO (CO-TPR)

All these experiments were performed on a lab-made reactor. The catalyst was first pretreated in a 5% H<sub>2</sub>/Ar atmosphere for 30 min, then cooled to room temperature, followed by purging with the Ar atmosphere. Subsequently, the atmosphere was switched to 2% CO/Ar, and the temperature was ramped up to 800 °C at a rate of 2 °C/min while simultaneously collecting data.

### In situ diffuse reflectance infrared Fourier transform spectroscopy (in situ DRIFTS)

In situ DRIFTS experiments was conducted using a Bruker Vertex 70 FTIR spectrometer equipped with a mercury-cadmium-telluride (MCT) detector. The sample cell (Harrick) featured CaF<sub>2</sub> windows and a heating cartridge. To enable rapid gas switching without dead volume, an electronically controlled quick-switching system was employed. About 20 mg of the sample was filled into the reaction cell and underwent pretreatment in a 5% H<sub>2</sub>/Ar atmosphere before any tests were conducted. The spectra were recorded with an integration time of 30 s and a resolution of 4 cm<sup>-1</sup>, while the background was taken in pure N<sub>2</sub>. For the CO adsorption experiment, after the H<sub>2</sub> pretreatment, the 2% CO/He gas was introduced into the reaction cell at 25 °C. Subsequently, the temperature was raised to 250 °C, and the 2% CO/10% H<sub>2</sub>O/Ar gas mixture was introduced into the system. After the WGS reaction, the temperature was lowered to room temperature, and 2% CO/He gas was reintroduced. For the CO adsorption–desorption experiments at 250 °C, the catalyst was exposed to 2% CO/He for 10 min, after which it was purged with N<sub>2</sub> and a mixture of N<sub>2</sub> and H<sub>2</sub>O, respectively. Upon reaching a new temperature, a fresh background measurement was taken in pure N<sub>2</sub>.

### In-situ infrared spectroscopy in the transmission mode

In-situ infrared spectroscopy was carried out in a UHV apparatus by an FTIR spectrometer Bruker Vertex 70 (v) with a multichambered ultrahigh vacuum system. The catalyst underwent four different pre-treatment methods. Firstly, the sample was pretreated with H<sub>2</sub> in the sample cell at 300 °C and dropped to –130 °C directly. Secondly, the sample was pretreated in the sample cell with H<sub>2</sub> at 300 °C, followed by the switch to CO atmosphere for further treatment, and then cooled down to –130 °C. Thirdly, the sample underwent the WGS reaction treatment in the reactor, then it was transferred to the sample cell and directly cooled down to –130 °C. Lastly, the sample underwent the WGS reaction treatment in the reactor, then it was transferred to the sample cell and further treated with CO at 300 °C, followed by cooling down to –130 °C. After the completion of the aforementioned pretreatments and cooling down to –130 °C, adsorption was performed by passing 1 × 10<sup>-1</sup> mbar CO accompanied by data collection.

### X-ray photoelectron spectroscopy (XPS)

The XPS measurements were carried out on a Thermo Scientific ESCALAB Xi+ XPS spectrometer (Thermo Fisher) with Al K $\alpha$  radiation (15 kV, 20 mA). Each spectrum was energy-calibrated using the C 1s signal at 284.8 eV as a reference.

### Raman test

All the ex situ and in situ Raman spectra were acquired by using a Raman microscope system (HORIBA JY) with laser excitation at 473 nm. In the in-situ Raman experiment, the sample was first pretreated with H<sub>2</sub> at 300 °C for 30 min, during which data was collected. Subsequently, the atmosphere was switched to CO, and after maintaining the CO atmosphere for another 30 minutes, data collection was performed.

### Catalytic tests

The WGS reaction of platinum-titanium catalysts was conducted in a fixed-bed reactor. 100 mg of the samples (20–40 mesh) was placed inside and pretreated with a 5% H<sub>2</sub>/Ar gas mixture (70 mL·min<sup>-1</sup>) at 300 °C for 30 min. Following this, a mixed gas (2% CO/10% H<sub>2</sub>O/Ar) was flowed through to carry out the reaction at a gas hourly space velocity of 42,000 mL·g<sub>cat</sub><sup>-1</sup>·h<sup>-1</sup>. The outlet gas compositions of CO and CO<sub>2</sub> were analyzed online on a Gasboard 3500 IR spectroscopy (Wuhan Sifang Company, China). The CO conversion was defined by the following formula:

$$X_{\text{CO}} = \frac{n_{\text{CO}}^{\text{in}} - n_{\text{CO}}^{\text{out}}}{n_{\text{CO}}^{\text{in}}} \times 100 \quad (1)$$

The reaction rates were evaluated by varying the catalyst mass and the flow rate of the reactant gas under conditions where the CO conversion remained below 15%, which were calculated by the following formula:

$$r = \frac{F \times X_{\text{CO}}}{m} \quad (2)$$

where  $F$  was the molar flow rate of CO in mol·h<sup>-1</sup> and  $m$  was the mass of Pt in the fixed bed.

The TOF value was defined by the following formula:

$$\text{TOF} = \frac{r \times M_{\text{Pt}}}{D_{\text{Pt}}} \quad (3)$$

where  $M_{\text{Pt}}$  was the molar weight of Pt and  $D_{\text{Pt}}$  was the dispersion of Pt calculated from the results of CO chemisorption by assuming that the stoichiometric ratio of adsorbed CO/Pt is 1.

During the stability test, the gas mixture was continuously supplied to the fixed-bed reactor at 250 °C for 70 h. Simultaneously, the conversion data were continuously monitored at gas hourly space velocities of 84,000 mL·g<sub>cat</sub><sup>-1</sup>·h<sup>-1</sup>. Moreover, the conversion data during the 70 h process of supplying gas to a fixed-bed reactor while alternating between 250 °C and 300 °C was also recorded. An ice trap was employed to condense the unreacted water following the reactor tube. The concentrations of CO and CO<sub>2</sub> in the outlet gas were monitored using an infrared gas analyzer (Gasboard-3100, manufactured by Wuhan Sifang Company, China). The apparent activation energy (*E<sub>a</sub>*) was determined using the same apparatus as for testing the catalytic performance mentioned earlier. A suitable quantity of Pt-supported catalyst was loaded into the reactor, and the temperature and gas flow rate were adjusted to maintain the CO conversion within the range of 5 to 15%. The outlet gas was then analyzed with an infrared gas analyzer (Gasboard-3100, Wuhan Sifang Company, China).

## Data availability

The main data supporting the findings of this study are available within the article and its Supplementary Information. Source data of Fig. 1b–e, 2, 3 and 4b–d are provided with this paper. Data are available from the corresponding author upon request. Source data are provided with this paper.

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## Author contributions

C.-J. Jia supervised the work; C.-X. Wang, W.-W. Wang and X.-P. Fu designed the experiments; C.-X. Wang analyzed the results and wrote the manuscript; C.-X. Wang and W.-W. Wang performed the in situ DRIFTS, Raman and XPS; and C.-X. Wang performed the catalysts preparation, tests, and characterization; F. R. Wang and H.-X. Liu provided the XAS measurements and data analysis.

## Competing interests

The authors declare no competing interests.

## Additional information

**Supplementary information** The online version contains supplementary material available at [\[redacted\]](#)

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